

Homework 4 Review

STAT 220B – Discussion

What HW4 was about

HW4 covered the last block of the course: invariance (Lecture 11) and the Stein/SURE machinery (Lectures 9 and 10). The two problems are each an *application* of a tool, taken to a setting the lectures did not work out.

- ▶ **Problem 1:** the **Pitman estimator** for a location family the lectures never touched, the Uniform. Lecture 11 derived Pitman for the Normal (giving $\bar{X}$); here the same machinery gives the **midrange**, and the midrange turns out to crush the sample mean.
- ▶ **Problem 2:** **SURE** applied to **soft thresholding** rather than to James-Stein. Lecture 9 mentioned SureShrink in passing; this problem derives the SureShrink risk formula from scratch and examines its behavior at the origin.

Both problems share a theme from the whole course: the “obvious” estimator (sample mean, or no shrinkage) is rarely the best one, and the right tool reveals exactly how much better you can do.

Problem 1: Pitman for the Uniform location family

The question

$X_1, \dots, X_n$ iid from the Uniform location family

$$f(x \mid \theta) = \mathbf{1}\{\theta - 1/2 \leq x \leq \theta + 1/2\}, \quad \theta \in \mathbb{R},$$

squared-error loss. The problem is invariant under the location group $\mathcal{G} = \{g_c : g_c(x) = x + c\}$.

(a) Show the joint density is $\prod_i f(x_i \mid \theta) = \mathbf{1}\{x_{(n)} - 1/2 \leq \theta \leq x_{(1)} + 1/2\}$. Then derive the Pitman estimator and show it equals the midrange $\frac{1}{2}(X_{(1)} + X_{(n)})$.

(b) Compute the risk of the midrange in closed form, compute the risk of $\bar{X}$, and show the ratio $R(\theta, \delta_P)/R(\theta, \bar{X})$ is free of θ and tends to 0 as $n \rightarrow \infty$.

The joint density

Each factor $f(x_i | \theta) = 1$ exactly when $\theta - 1/2 \leq x_i \leq \theta + 1/2$. Flip this to a statement about θ :

$$f(x_i | \theta) = 1 \iff x_i - 1/2 \leq \theta \leq x_i + 1/2.$$

The product is 1 only when *every* such two-sided constraint holds, i.e. on the intersection of all n intervals:

$$\prod_{i=1}^n f(x_i | \theta) = \mathbf{1}\left\{\max_i(x_i - 1/2) \leq \theta \leq \min_i(x_i + 1/2)\right\} = \mathbf{1}\{x_{(n)} - 1/2 \leq \theta \leq x_{(1)} + 1/2\}.$$

The interval is nonempty because all x_i lie within a window of width 1, so $x_{(n)} - x_{(1)} \leq 1$.

Teaching point. The likelihood is *flat* on this interval (it equals 1 throughout, 0 outside). The data localize θ to a box but say nothing about where in the box. This is what makes the MLE non-unique and what the Pitman machinery resolves.

The Pitman estimator

Under squared-error loss, the Pitman estimator is the generalized Bayes posterior mean against the flat prior $\pi(\theta) = 1$ (Lecture 11, Berger §6.6). With $L = x_{(n)} - 1/2$ and $U = x_{(1)} + 1/2$, the posterior is uniform on $[L, U]$, so the posterior mean is the midpoint:

$$\delta_P(x) = \frac{\int_L^U \theta d\theta}{\int_L^U d\theta} = \frac{(U^2 - L^2)/2}{U - L} = \frac{U + L}{2} = \frac{(x_{(1)} + 1/2) + (x_{(n)} - 1/2)}{2} = \frac{x_{(1)} + x_{(n)}}{2}.$$

So the best equivariant estimator is the **midrange**: average the smallest and largest observations. The $\pm 1/2$ offsets cancel exactly, which is why the answer is clean.

Teaching point. Compare with the Normal location problem from Lecture 11, where the same construction gave $\bar{X}$. The Pitman recipe is identical; only the likelihood shape changed. Different models put their information in different statistics: the Normal in the mean, the Uniform in the extremes.

Setting up the risk computation

By Theorem 1 of Lecture 11, the risk of any equivariant rule is constant in θ , so compute at $\theta = 0$. Set $Y_i = X_i - \theta \sim \text{Uniform}(-1/2, 1/2)$ iid. Then

$$\delta_P - \theta = \frac{Y_{(1)} + Y_{(n)}}{2}, \quad \bar{X} - \theta = \bar{Y}.$$

By symmetry of $\text{Uniform}(-1/2, 1/2)$ about 0, $\mathbb{E}[Y_{(1)} + Y_{(n)}] = 0$. So the midrange is unbiased and its risk is pure variance:

$$R(\theta, \delta_P) = \mathbb{E} \left[\left(\frac{Y_{(1)} + Y_{(n)}}{2} \right)^2 \right] = \frac{1}{4} \text{Var}(Y_{(1)} + Y_{(n)}).$$

Risk of the midrange

Shift to $U_i = Y_i + 1/2 \sim \text{Uniform}(0, 1)$, which leaves the variance unchanged:
 $\text{Var}(Y_{(1)} + Y_{(n)}) = \text{Var}(U_{(1)} + U_{(n)})$. Using the given order-statistic moments,

$$\text{Var}(U_{(1)} + U_{(n)}) = \text{Var}(U_{(1)}) + \text{Var}(U_{(n)}) + 2 \text{Cov}(U_{(1)}, U_{(n)}) = \frac{2n+2}{(n+1)^2(n+2)} = \frac{2}{(n+1)(n+2)}.$$

Therefore

$$R(\theta, \delta_P) = \frac{1}{4} \cdot \frac{2}{(n+1)(n+2)} = \frac{1}{2(n+1)(n+2)}.$$

Teaching point. Note the positive covariance between min and max: when the whole sample shifts up, both $U_{(1)}$ and $U_{(n)}$ rise together. That positive correlation is what makes the midrange's variance shrink like $1/n^2$ rather than $1/n$.

Comparison with the sample mean

The sample mean has the textbook rate. With $\text{Var}(Y_i) = 1/12$ (variance of a unit-width uniform),

$$R(\theta, \bar{X}) = \text{Var}(\bar{Y}) = \frac{1}{12n}.$$

The ratio:

$$\frac{R(\theta, \delta_P)}{R(\theta, \bar{X})} = \frac{1/[2(n+1)(n+2)]}{1/(12n)} = \frac{6n}{(n+1)(n+2)} \sim \frac{6}{n} \rightarrow 0.$$

The ratio is free of θ (both risks are constant by Theorem 1), and the midrange is asymptotically infinitely better: its risk is order $1/n^2$ versus the sample mean's $1/n$.

Teaching point. For the Uniform, the extremes are nearly sufficient and the sample mean throws away most of the information. The lesson is the mirror image of the Normal case: there, $\bar{X}$ is optimal and the midrange is terrible (heavy-tailed, no finite-sample efficiency). The “right” estimator is dictated by the model, not by habit.

Problem 2: SURE for soft thresholding

The question

$\mathbf{X} \sim \mathcal{N}_p(\theta, I_p)$, squared-error loss. The soft-thresholding estimator is

$$\delta^\lambda(\mathbf{x})_i = \text{sign}(x_i)(|x_i| - \lambda)^+,$$

and $\delta^\lambda(\mathbf{x}) = \mathbf{x} + \mathbf{g}(\mathbf{x})$ with $g_i(\mathbf{x}) = -\text{sign}(x_i) \min(|x_i|, \lambda)$.

(a) Compute $\partial g_i / \partial x_i$, then $\nabla \cdot \mathbf{g}$ and $\|\mathbf{g}\|^2$.

(b) Apply SURE to get $\hat{R}^\lambda(\mathbf{x}) = p - 2 \#\{i : |x_i| \leq \lambda\} + \sum_i \min(x_i^2, \lambda^2)$.

(c) At $\theta = \mathbf{0}$, derive $R(\mathbf{0}, \delta^\lambda) = 2p[(1 - \Phi(\lambda))(1 + \lambda^2) - \lambda\phi(\lambda)]$ and show it tends to 0 as $\lambda \rightarrow \infty$.

The derivative of the correction term

The function $g_i(\mathbf{x}) = -\text{sign}(x_i) \min(|x_i|, \lambda)$ is piecewise. Look at the two regions:

- ▶ **Outside the threshold**, $|x_i| > \lambda$: here $\min(|x_i|, \lambda) = \lambda$, so $g_i = -\lambda \text{sign}(x_i)$ is *constant* in x_i . Hence $\partial g_i / \partial x_i = 0$.
- ▶ **Inside the threshold**, $|x_i| < \lambda$: here $\min(|x_i|, \lambda) = |x_i|$, so $g_i = -|x_i| \text{sign}(x_i) = -x_i$. Hence $\partial g_i / \partial x_i = -1$.

For $j \neq i$, g_i does not depend on x_j . Compactly, almost everywhere,

$$\frac{\partial g_i}{\partial x_i}(\mathbf{x}) = -\mathbf{1}\{|x_i| < \lambda\}.$$

Teaching point. The derivative is exactly the indicator that coordinate i got thresholded to zero. The geometry of soft thresholding is: leave large coordinates alone (derivative 0 in the correction), kill small ones (correction cancels x_i entirely).

Divergence and norm

Summing the diagonal derivatives gives the divergence, which counts the thresholded coordinates:

$$\nabla \cdot \mathbf{g}(\mathbf{x}) = \sum_{i=1}^p \frac{\partial g_i}{\partial x_i}(\mathbf{x}) = -\#\{i : |x_i| < \lambda\}.$$

(The boundary $\{|x_i| = \lambda\}$ has measure zero, so $<$ and $\leq$ agree under expectation.)

The squared norm sums the squared corrections:

$$\|\mathbf{g}(\mathbf{x})\|^2 = \sum_{i=1}^p g_i(\mathbf{x})^2 = \sum_{i=1}^p (\min(|x_i|, \lambda))^2 = \sum_{i=1}^p \min(x_i^2, \lambda^2).$$

A coordinate contributes x_i^2 if it was thresholded (the whole observation is the error) and λ^2 if it survived (the shift is exactly λ).

The SURE formula

Plug into the SURE template from Lecture 9, $\hat{R} = p + 2\nabla \cdot \mathbf{g} + \|\mathbf{g}\|^2$:

$$\hat{R}^\lambda(\mathbf{x}) = p - 2 \#\{i : |x_i| \leq \lambda\} + \sum_{i=1}^p \min(x_i^2, \lambda^2).$$

By Stein's identity applied coordinatewise, $\mathbb{E}_\theta[\hat{R}^\lambda(\mathbf{X})] = R(\theta, \delta^\lambda)$ for every θ .

Teaching point. This is the Donoho-Johnstone SureShrink criterion. Every term is computable from the data alone, with no knowledge of θ . That is exactly what makes SURE useful for *choosing* λ in practice: minimize $\hat{R}^\lambda(\mathbf{x})$ over λ and you approximate the oracle threshold without knowing the truth.

Risk at the origin

Take $\theta = \mathbf{0}$, so $X_i \sim \mathcal{N}(0, 1)$ iid. By unbiasedness of SURE, $R(\mathbf{0}, \delta^\lambda) = \mathbb{E}_{\mathbf{0}}[\hat{R}^\lambda(\mathbf{X})]$.
Two expectations, both per-coordinate times p :

Count term. $\mathbb{E}[\mathbf{1}\{|X_i| \leq \lambda\}] = \mathbb{P}(|Z| \leq \lambda) = 2\Phi(\lambda) - 1$.

Min term. Split $\min(Z^2, \lambda^2) = Z^2\mathbf{1}\{|Z| \leq \lambda\} + \lambda^2\mathbf{1}\{|Z| > \lambda\}$. The hint gives $\mathbb{E}[Z^2\mathbf{1}\{|Z| \leq \lambda\}] = 2\Phi(\lambda) - 1 - 2\lambda\phi(\lambda)$, and $\mathbb{P}(|Z| > \lambda) = 2(1 - \Phi(\lambda))$, so

$$\mathbb{E}[\min(Z^2, \lambda^2)] = 2\Phi(\lambda) - 1 - 2\lambda\phi(\lambda) + 2\lambda^2(1 - \Phi(\lambda)).$$

Assembling the closed form

$$\begin{aligned}R(\mathbf{0}, \delta^\lambda) &= p - 2p(2\Phi(\lambda) - 1) + p[2\Phi(\lambda) - 1 - 2\lambda\phi(\lambda) + 2\lambda^2(1 - \Phi(\lambda))] \\&= p[1 - 4\Phi(\lambda) + 2 + 2\Phi(\lambda) - 1 - 2\lambda\phi(\lambda) + 2\lambda^2(1 - \Phi(\lambda))] \\&= p[2 - 2\Phi(\lambda) - 2\lambda\phi(\lambda) + 2\lambda^2(1 - \Phi(\lambda))] \\&= 2p[(1 - \Phi(\lambda))(1 + \lambda^2) - \lambda\phi(\lambda)].\end{aligned}$$

Limit. Mill's ratio gives $1 - \Phi(\lambda) \sim \phi(\lambda)/\lambda$, so $(1 - \Phi(\lambda))(1 + \lambda^2) \sim \lambda\phi(\lambda)$ and the bracket is $o(\lambda\phi(\lambda))$. Since $\lambda\phi(\lambda) \rightarrow 0$,

$$R(\mathbf{0}, \delta^\lambda) \rightarrow 0 \quad \text{as } \lambda \rightarrow \infty.$$

What the origin result does and does not say

At $\theta = \mathbf{0}$, pushing $\lambda \rightarrow \infty$ drives the risk to 0. Every coordinate gets thresholded to zero, and zero is exactly the truth, so the error vanishes.

This is **not** a violation of the minimax bound $\inf_{\delta} \sup_{\theta} R(\theta, \delta) = p$. That bound is about the *worst case over θ* . At the single point $\theta = \mathbf{0}$, any rule can do arbitrarily well, and the constant rule $\delta \equiv \mathbf{0}$ does perfectly. The price is paid elsewhere: for θ far from $\mathbf{0}$, large- λ thresholding has risk close to $\|\theta\|^2$, which blows up. So $\sup_{\theta} R(\theta, \delta^{\lambda}) \rightarrow \infty$ as $\lambda \rightarrow \infty$, fully consistent with minimaxity.

Teaching point. This is the same moral as HW3 Problem 4 (restricted-mean minimax): a rule can be spectacular at a favored point and disastrous in the worst case. Minimax cares only about the worst case; SURE lets you *adapt* λ to the data so you do well wherever θ actually is.

Wrap-up

The two problems in one frame

	Problem 1	Problem 2
Tool	Pitman / invariance (L11)	SURE (L9, L10)
Lecture did	Normal $\rightarrow \bar{X}$	James-Stein
HW did	Uniform $\rightarrow$ midrange	Soft thresholding
Key result	$R(\delta_P)/R(\bar{X}) \rightarrow 0$	$\hat{R}^\lambda = \text{SureShrink}$
Punchline	model picks the statistic	risk estimable from data

Two recurring lessons from the whole course show up again here.

The obvious estimator is rarely optimal. For the Uniform, the sample mean is asymptotically infinitely worse than the midrange. For the Normal mean in high dimensions, no shrinkage is inadmissible. The right estimator depends on the model and the loss, never on habit.

Worst-case and pointwise behavior are different questions. The soft-threshold rule with large λ is perfect at the origin and terrible far away. Minimax weighs only the worst case: SURE gives a data-driven way to chase the best λ for the θ you actually